EE-312L Signals and Systems Lab
Laboratory Session 12
To Understand and Analyze the Frequency Response of Continuous-Time and Discrete-Time Systems using MATLAB

January 1, 2025

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1 Objective

The objective of this lab session is to understand and analyze the frequency response of continuous-time and discrete-time systems using MATLAB.

2 Frequency Response of Continuous-Time Systems

The frequency response of a system is expressed as

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{\sum_{k=0}^{M} b_k (j\omega)^k}{\sum_{k=0}^{N} a_k (j\omega)^k} \tag{1}$$

The MATLAB command freqs is used to compute and plot the frequency response of continuous-time systems.

Example 01:

Plot the magnitude and the phase of the frequency response

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{8(j\omega)^2 + 2(j\omega) + 20}{6(j\omega)^2 - 5(j\omega) - 10} \tag{2}$$

in the frequency interval $0 \leq \omega \leq 10$ rad/s.

```

clc;
clear all;
```

```
close all;

num = [8 2 20];
den = [6 -5 -10];
w = 0:.1:10;

H = freqs(num,den,w)


subplot(2,1,1)
plot(w,abs(H), 'linewidth', 2);
legend('Magnitude of H(e^{\omega}), 0<\omega<10')

subplot(2,1,2)
plot(w,angle(H), 'r','linewidth', 2);
legend(' Phase of H(e^{\omega}),0<\omega <10')
```

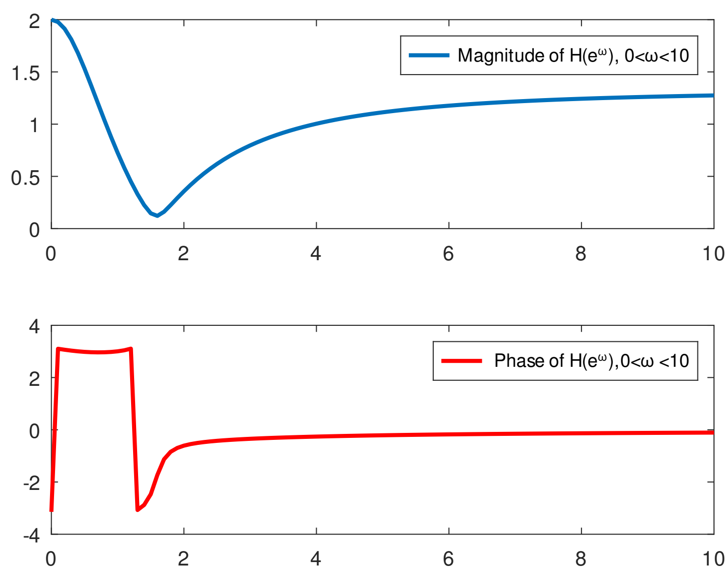

Figure 1: Magnitude and phase response of a continuous-time system

3 Frequency Response of Discrete-Time Systems

The frequency response of a system is expressed as

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{\sum_{k=0}^{M} b_k e^{-jk\omega}}{\sum_{k=0}^{N} a_k e^{-jk\omega}} \tag{3}$$

The MATLAB command freqz is used to compute and plot the frequency response of discrete-time systems.

Example 02:

Plot the magnitude and the phase of the frequency response

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{3 + 5e^{-j\omega} - 7e^{-2j\omega}}{2 - 4e^{-j\omega}} \tag{4}$$

in the frequency interval $0 \leq \omega \leq 2\pi$ rad/s.

```

clc;
clear all;
close all;

w = 0:.1:2*pi;
num = [3 5 -7];
den = [2 -4];

H = freqz(num,den,w)


subplot(2,1,1)
plot(w,abs(H), 'linewidth', 2);
legend('|H(e^{j\omega})|')
xlim([0 2*pi])

subplot(2,1,2)
plot(w,angle(H), 'r','linewidth', 2);
xlim([0 2*pi])
legend('\angle H(e^{j\omega})')
```

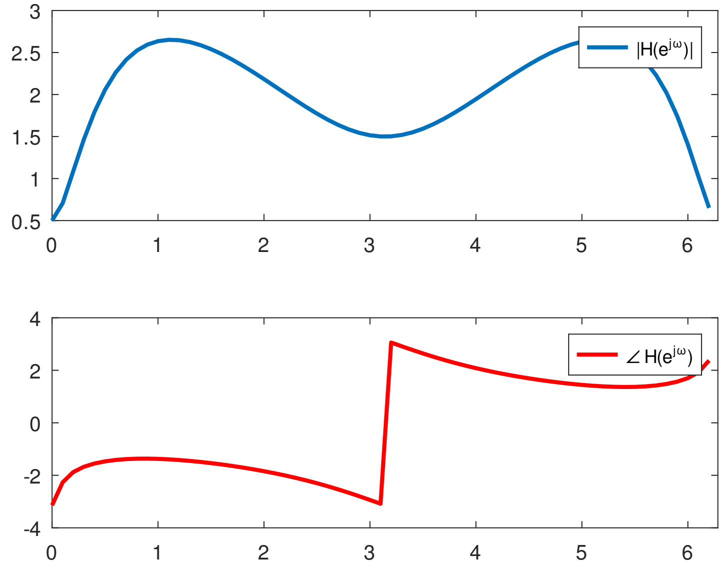

Figure 2: Magnitude and phase response of a discrete-time system

4 Lab 12: Lab Tasks

1. Plot the magnitude and the phase of the frequency response of of the continuous-time system

$$H(j\omega) = \frac{8(j\omega)^2 + 2(j\omega) + 9}{6(j\omega)^2 - 10} \tag{5}$$

in the frequency interval $0 \leq \omega \leq 10$ rad/s.

2. Plot the magnitude and the phase of the frequency response of of the discrete-time system

$$H(e^{j\omega}) = \frac{2 + 5e^{-j\omega} - 7e^{-2j\omega}}{1 - 4e^{-j\omega}} \tag{6}$$

References

[PV10] Alex Palamides and Anastasia Veloni. *Signals and systems laboratory with MATLAB*. CRC press, 2010.